- A function $g(t)$ is called $o(t)$ if it satisfies
$$\lim_{t \to 0} g(t)/t = 0$$

- Implication of Taylor's theorem:

$$f(y) = f(x) + \nabla f^t(x)\,(y-x) + o(\|y-x\|)$$

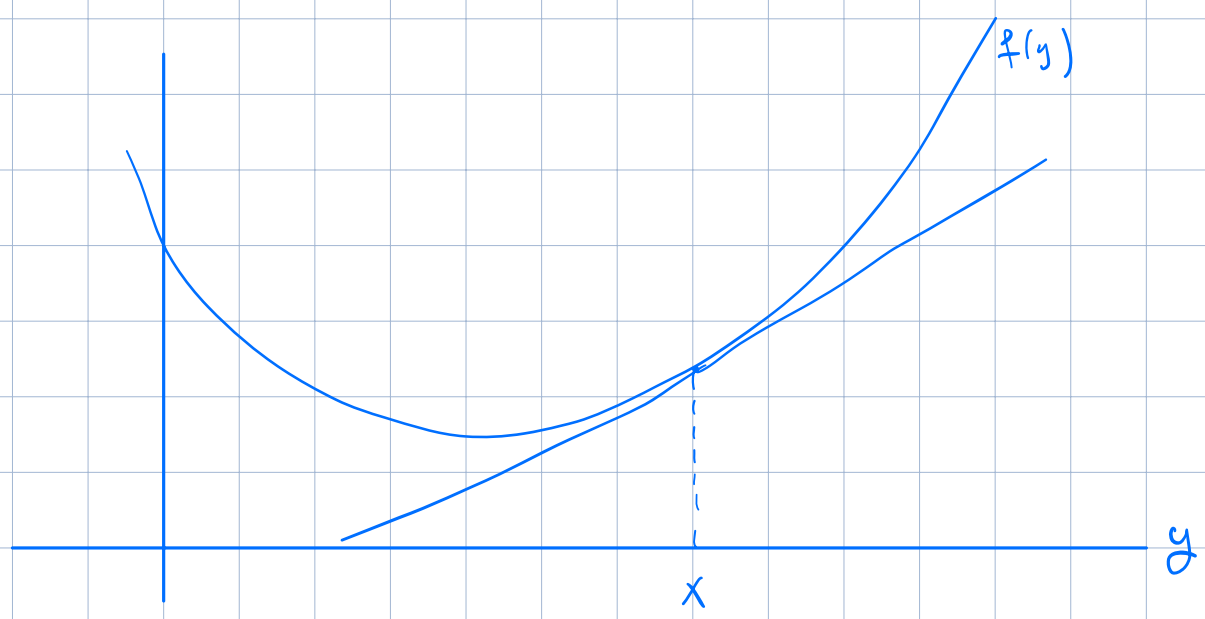
- Definition of a convex function:
$f: \mathbb{R}^n \to \mathbb{R}$ is convex if and only if

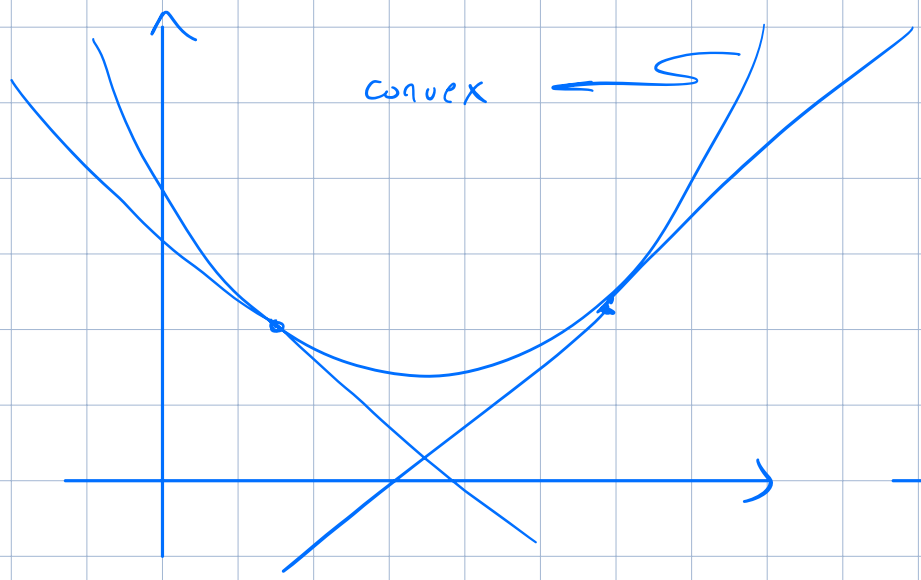
$$f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$$

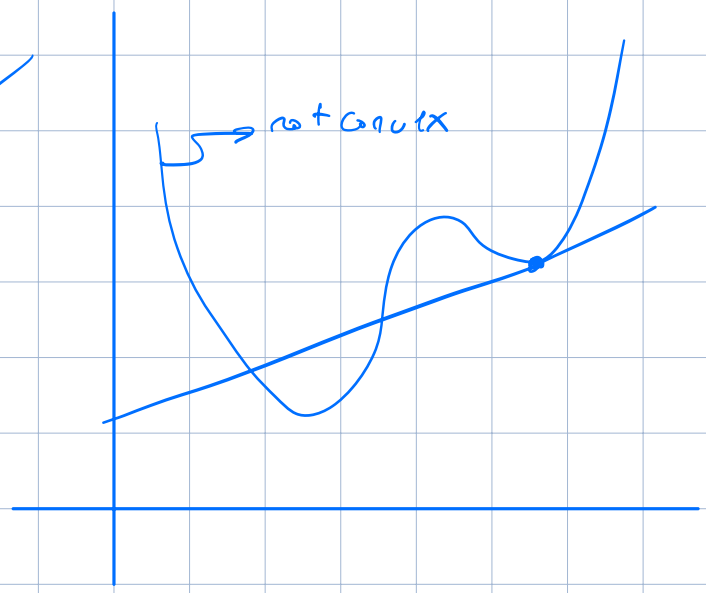
First-order characterization of convexity:

- The function $f: \mathbb{R}^n \to \mathbb{R}$ is convex if and only if

$$f(y) \geq f(x) + \nabla f^t(x) \cdot (y-x) \quad \text{for all } x, y \in \mathbb{R}^n.$$







<u>Theorem:</u> The function $f: \mathbb{R}^n \to \mathbb{R}$ is convex if and only if

$$f(y) \geq f(x) + Df^T(x)\,(y-x) \quad \text{for all } x, y \in \mathbb{R}^n$$

<u>Pf:</u> $\Rightarrow$ If f is convex, then $f(y) \geq f(x) + \nabla f(x)\,(y-x)$ for all $x, y \in \mathbb{R}^n$.

$$\cancel{f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)}$$

Because f is convex

$$f(x + \lambda(y-x)) = f((1-\lambda)x + \lambda y) \leq (1-\lambda) f(x) + \lambda f(y)$$

$$\Rightarrow f(x + \lambda(y-x)) \leq (1-\lambda) f(x) + \lambda f(y)$$

$$\Rightarrow \lambda f(y) \geq \lambda f(x) + f(x + \lambda(y-x)) - f(x)$$

$$\Rightarrow f(y) \geq f(x) + \frac{f(x + \lambda(y-x)) - f(x)}{\lambda}$$

$$= f(x) + \frac{D^t f(x) \cdot (\cancel{x} + \lambda(y-x) - \cancel{x}) + o(\|\cancel{x} + \lambda(y-x) - \cancel{x}\|)}{\lambda}$$

implication of taylor's theorem

$$= f(x) + \frac{\lambda\, D f^t(x) \cdot (y-x) + o(\lambda \|y-x\|)}{\lambda}$$

$$= f(x) + D f^t(x)\,(y-x) + \frac{o(\lambda \|y-x\|)}{\lambda}$$

$$= f(x) + D^t f(x)\,(y-x) + \frac{o(\lambda \|y-x\|)}{\lambda \|y-x\|} \|y-x\| \qquad (\ddagger)$$

holds for all x, y and λ

Note $\displaystyle\lim_{\lambda \to 0} \frac{o(\lambda \|y-x\|)}{\lambda \|y-x\|} = 0$

By $(\ddagger)$, for all $x, y \in \mathbb{R}^n$, $\lambda \in [0,1]$, we have

$$f(y) \geq f(x) + D f^t(x)\,(y-x) + \frac{o(\lambda \|y-x\|)}{\lambda \|y-x\|} \cdot \|y-x\|$$

So,

$$f(y) \geq f(x) + D f^t(\cancel{x})\,(y-x) + \underbrace{\cancel{\lim_{\lambda \to 0} \frac{o(\lambda \|y-x\|)}{\lambda \|y-x\|} \|y-x\|}}_{0}$$

done!

$\Leftarrow$ If $f(y) \geq f(x) + D f^t(x) \cdot (y-x)$ for all $x, y \in \mathbb{R}^n$, then f is convex

Take any $a, b \in \mathbb{R}^n$, and any $\lambda \in [0,1]$

$$f(a) \geq f(\lambda a + (1-\lambda) b) + Df^t(\lambda a + (1-\lambda) b) \cdot (a - (\lambda a + (1-\lambda) b))$$
$$f(b) \geq f(\lambda a + (1-\lambda) b) + Df^t(\lambda a + (1-\lambda) b) \cdot (b - (\lambda a + (1-\lambda) b))$$

$$f(a) \geq f(\lambda a + (1-\lambda) b) + (1-\lambda) \cdot Df^t(\lambda a + (1-\lambda) b)(a-b) \quad * \lambda$$
$$+ \; f(b) \geq f(\lambda a + (1-\lambda) b) - \lambda \, Df^t(\lambda a + (1-\lambda) b)(a-b) \quad * (1-\lambda)$$

$$\lambda f(a) + (1-\lambda) f(b) \geq \lambda f(\lambda a + (1-\lambda) b) + (1-\lambda) f(\lambda a + (1-\lambda) b)$$
$$= f(\lambda a + (1-\lambda) b)$$

So we have $\lambda f(a) + \lambda f(b) \geq f(\lambda a + (1-\lambda) b)$ Done

Minimizing Convex Functions

Definition of a global minimum:
The point x^* is a global minimum of f if
$$f(x^*) \leq f(y) \quad \text{for all } y \in \mathbb{R}^n.$$

Definition of a local minimum:
The point $\hat{x}$ is a local minimum of f if there exists some $\varepsilon > 0$ such that
$$f(\hat{x}) \leq f(y) \quad \text{for all } y \text{ that}$$
satisfies $\|y - \hat{x}\| \leq \varepsilon$.

Theorem: Let $f: \mathbb{R}^n \to \mathbb{R}$ be convex and $\hat{x}$ be a local minimum of f. Then $\hat{x}$ has to be a global minimum of f.

Pf: Fix any $y \in \mathbb{R}^n$, we want to show that $f(\hat{x}) \leq f(y)$

Because $\hat{x}$ is a local minimum of f, there exists some $\varepsilon > 0$, such that $f(\hat{x}) \leq f(z)$ for any z that satisfies $\|z - \hat{x}\| \leq \varepsilon$.

Choose λ such that $(1-\lambda) \leq \varepsilon / \|\hat{x} - y\|$, where y is as fixed at the beginning of the proof.

$$\|\underbrace{\lambda \hat{x} + (1-\lambda) y}_{} - \underbrace{\hat{x}}_{}\| = \|(1-\lambda)y - (1-\lambda)\hat{x}\| = \|(1-\lambda)(y - \hat{x})\|$$
$$= (1-\lambda)\|y - \hat{x}\| \leq \varepsilon$$

Thus $\lambda \hat{x} - (1-\lambda) y$ is within ε-neighborhood of local min $\hat{x}$.

$$f(\hat{x}) \leq f(\lambda \hat{x} + (1-\lambda) y)$$

Thus, we get

$$f(\hat{x}) \leq f(\lambda \hat{x} + (1-\lambda) y) \leq \lambda f(\hat{x}) + (1-\lambda) f(y)$$

So $f(\hat{x}) \leq \lambda f(\hat{x}) + (1-\lambda) f(y)$

$$\Rightarrow \cancel{(1-\lambda)} f(\hat{x}) \leq \cancel{(1-\lambda)} f(y) \Rightarrow f(\hat{x}) \leq f(y)$$

Done!

Theorem: Let $f: \mathbb{R}^n \to \mathbb{R}$ be a convex function. If $\nabla f(\hat{x}) = 0$, then $\hat{x}$ is a global minimum of f.

Pf: Fix any point $y \in \mathbb{R}^n$, we want to show $f(\hat{x}) \leq f(y)$

By first order characterization of convexity:

$$f(y) \geq f(\hat{x}) + \underbrace{\cancel{\nabla f(\hat{x}) \cdot (y - \hat{x})}}_{0}$$

So we get $f(y) \geq f(\hat{x})$ Done!